

Streaming MIP Algorithm — Single-Pass Memory Efficiency



Peak memory: $O(H \cdot W)$ vs $O(D \cdot H \cdot W)$ for full volume load
→ ~20 MB regardless of volume depth ($D = 300\text{--}2000$ slices)

```
np.maximum(mip_axis0, arr, out=mip_axis0) ← in-place  $O(1)$  update
```